

Retail Customer Purchase & Sales Performance Analytics

1. Project Overview and Objective

The **Retail Customer Purchase & Sales Performance Analytics** project focuses on analyzing retail sales and customer transaction data to understand **sales performance, customer behavior, order patterns, profitability, and operational performance**.

The project uses data related to customers, products, orders, sales, profit, payment methods, order status, returns, customer satisfaction, and shipping time. The data is cleaned, transformed, analyzed, and visualized using **Python, SQL, Excel, and Power BI** to develop meaningful business insights.

2. Data Sources

- **Domain:** E-Commerce

3. Problem Statement

Retail businesses generate large volumes of customer and transaction data, but raw data alone does not provide a clear understanding of **sales performance, customer behavior, profitability, and operational efficiency**.

The key challenge is to analyze the available retail data and identify meaningful patterns related to:

- **Sales and profit performance** across different years, cities, and product categories.
- **Customer purchasing behavior** and demographic patterns.
- **Order status**, including delivered, pending, cancelled, and returned orders.
- **Product performance** and identification of top-selling products.
- **Payment method preferences** and their relationship with sales.
- **Customer satisfaction** across different product categories.
- **Shipping performance** and variations in delivery timelines.
- **Returns and cancellations** that may affect overall business performance.

Therefore, there is a need to **clean, analyze, and visualize the retail dataset** through an interactive dashboard that converts raw transactional data into **clear, actionable business insights** and supports data-driven decision-making.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
order_id	String	Unique identification number of the order.

order_date	Date	Date on which the order was placed.
customer_id	String	Unique identification number of the customer.
customer_name	String	Name of the customer.
Age	Integer	Age of the customer.
Gender	String	Gender of the customer.
Region	String	Geographical region associated with the order/customer.
City	String	City associated with the order/customer.
product_category	String	Category of the purchased product.
product_name	String	Name of the purchased product.
Quantity	Integer	Number of units purchased.
unit_price	Float	Price of one unit of the product.
discount_pct	Float	Discount percentage applied to the product/order.
sales_amount	Float	Total sales amount generated from the order.
Profit	Float	Profit generated from the order.
shipping_cost	Float	Cost incurred for shipping the order.
payment_method	String	Payment method used for the order.
customer_satisfaction	Integer	Customer satisfaction rating.
return_flag	Boolean	Indicates whether the order was returned (True/False).
order_status	String	Current status of the order, such as Delivered, Pending, Cancelled, Returned, or Shipped.
days_to_ship	Integer	Number of days taken to ship the order.

5. Tools & Technologies

- **Python:** Data cleaning, preprocessing, exploratory data analysis, and statistical analysis.
 - **Pandas:** Data manipulation, missing-value handling, data transformation, grouping, and aggregation.
 - **Matplotlib & Seaborn:** Data visualization, trend analysis, and exploratory charts.
 - **Power BI:** Data modelling, DAX calculations, visualization, KPI analysis, and interactive dashboard creation.
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6. Data Pre-Processing (Python/Power BI)

Tasks Performed:

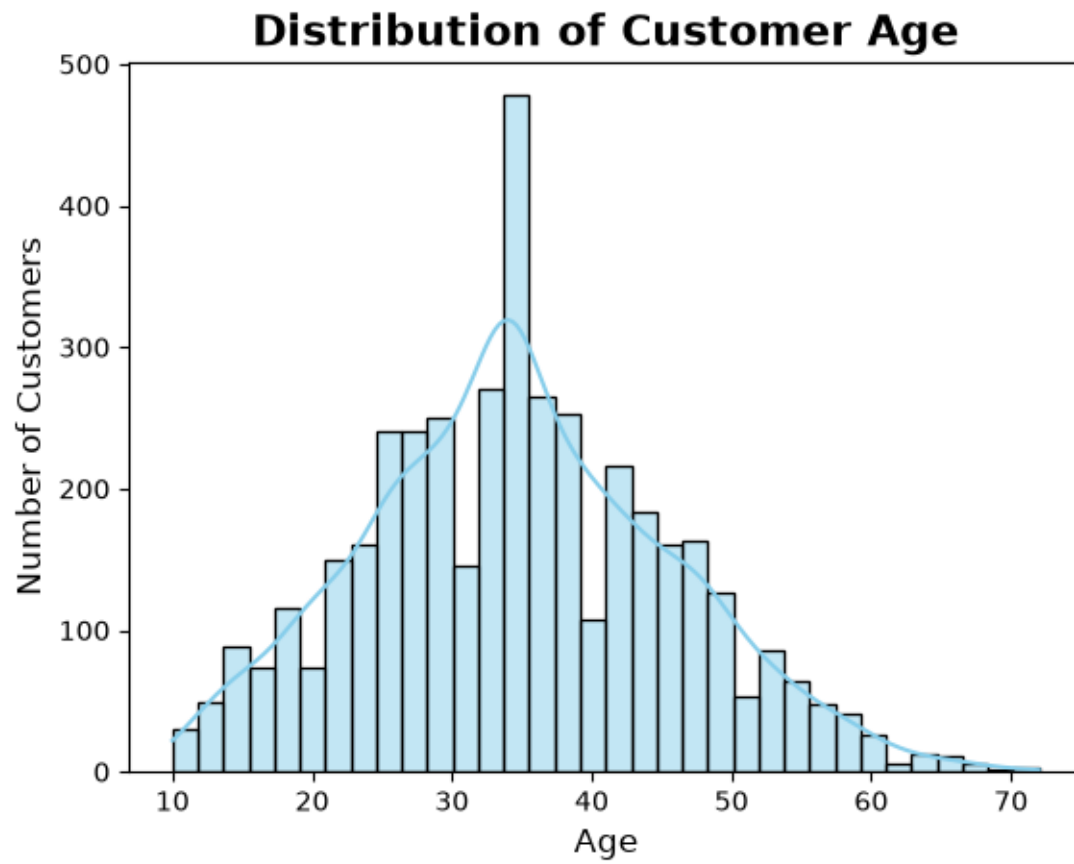
- **Data Cleaning & Transformation:**
Cleaned and transformed the dataset using Python, handled missing values and inconsistent data, removed duplicates, standardized data formats, and prepared the data for analysis.
 - **Data Modelling & Measures:** Created relationships, calculated columns, and DAX measures in Power BI to support analysis and dashboard development.
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7. Data Modelling and DAX (Power BI)

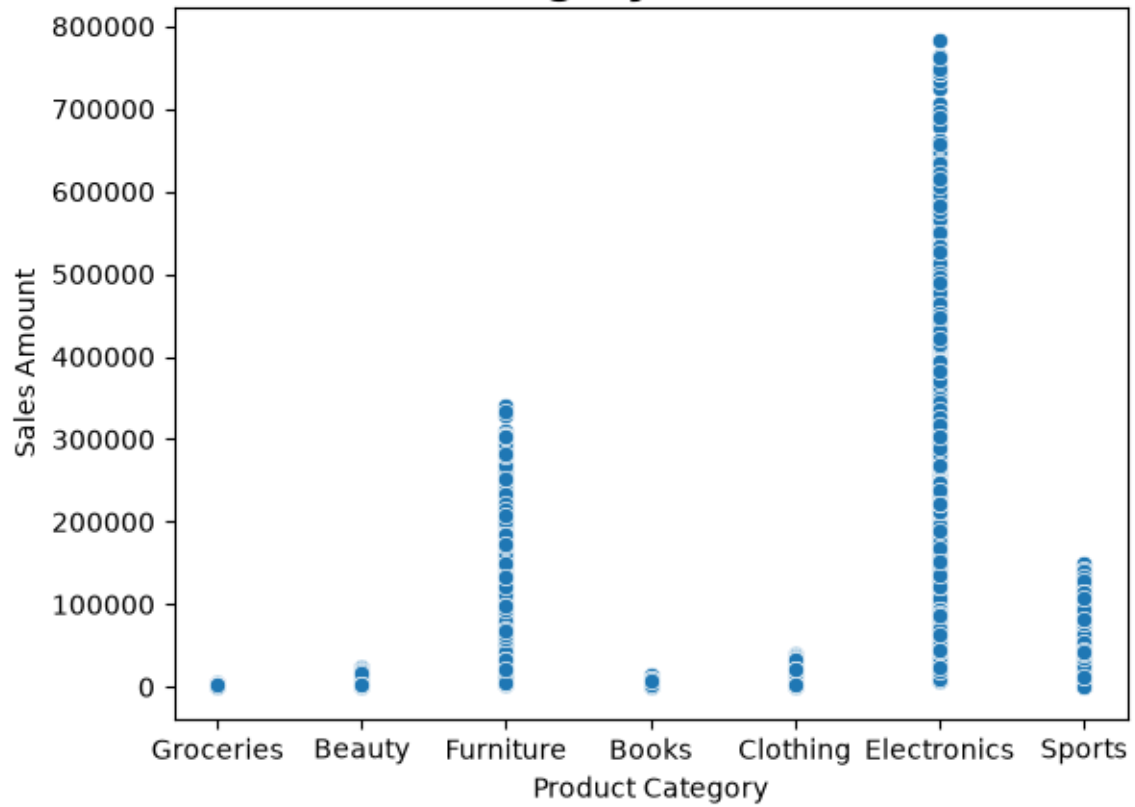
- **Calculated Columns & DAX Measures:** Created calculated columns and DAX measures to calculate key retail metrics, such as total sales, total profit, total orders, unique customers, average customer satisfaction

Customers	Orders	Total_Sales(Cr)	Total_Profit(Cr)	Avg_Satisfaction
4K	4.2K	25.38	4.95	3.02

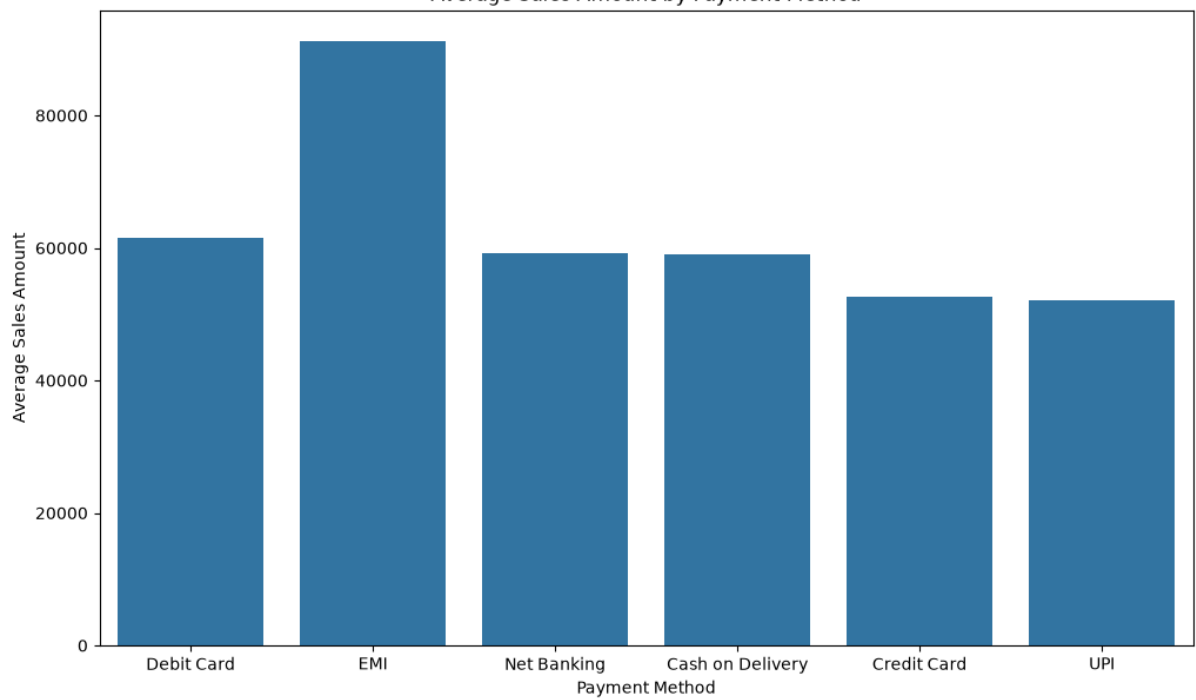
8. Analysis and Visualizations (Power BI)

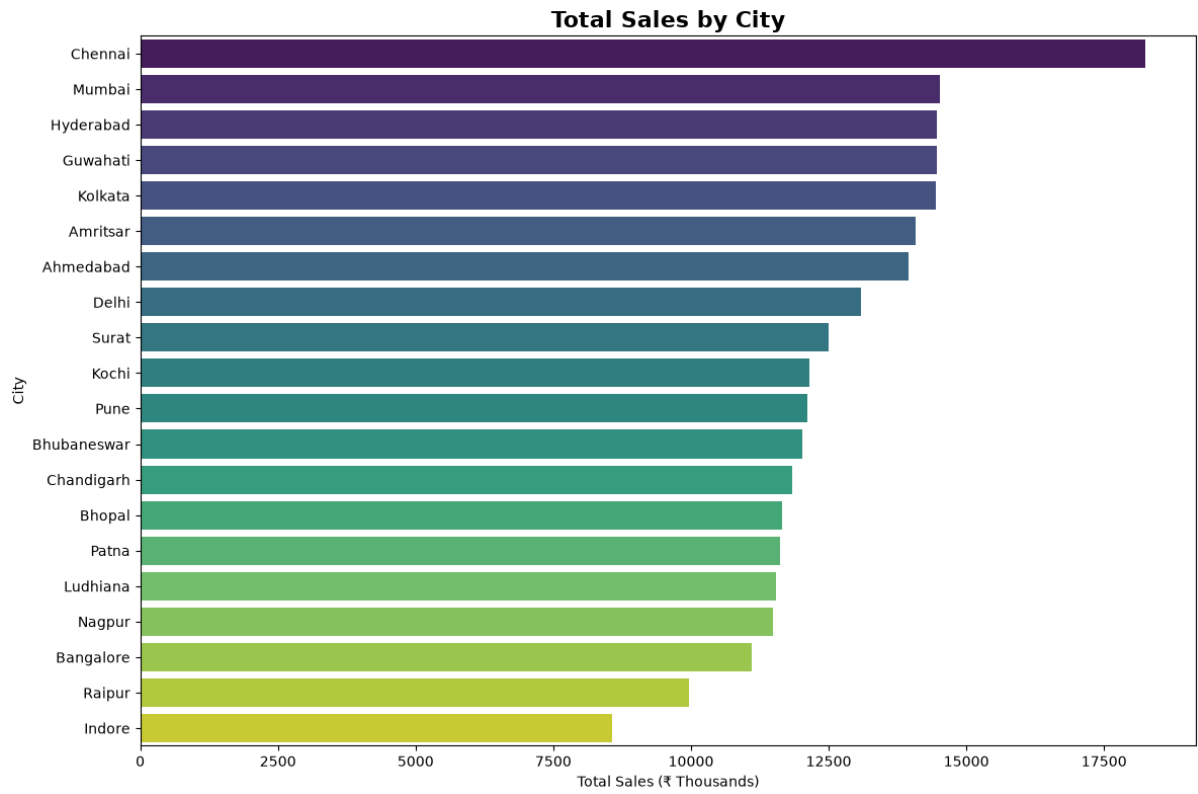
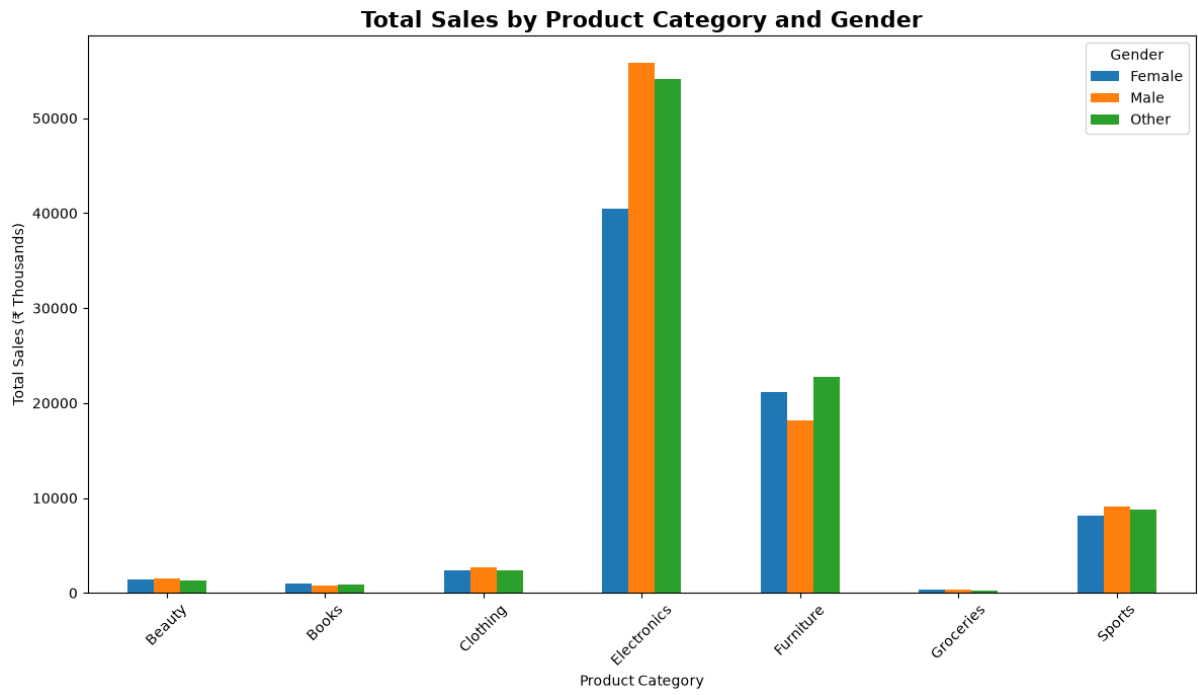


Product Category vs Sales Amount

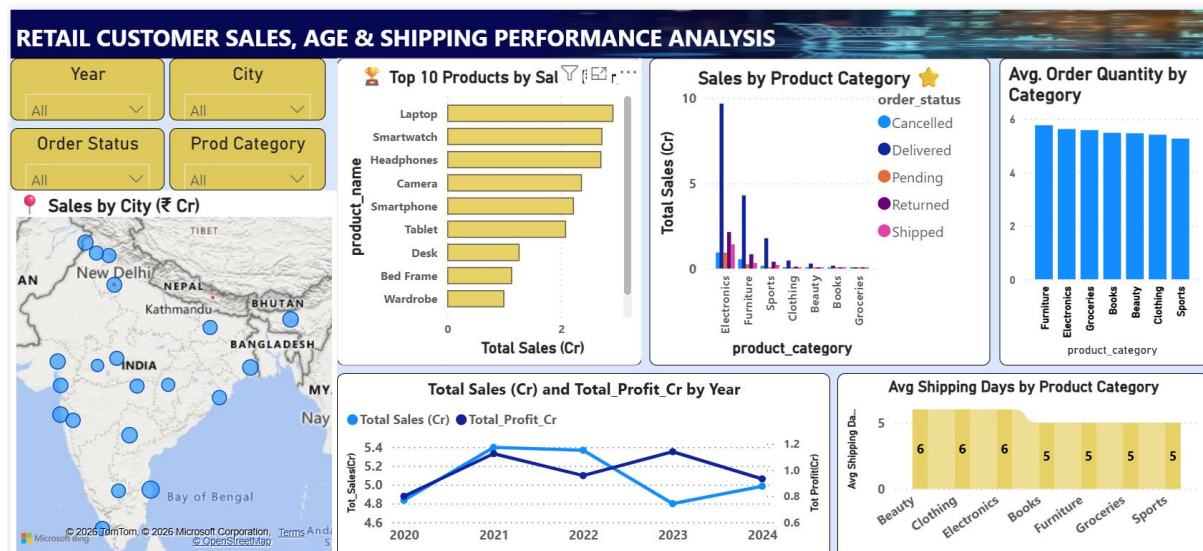
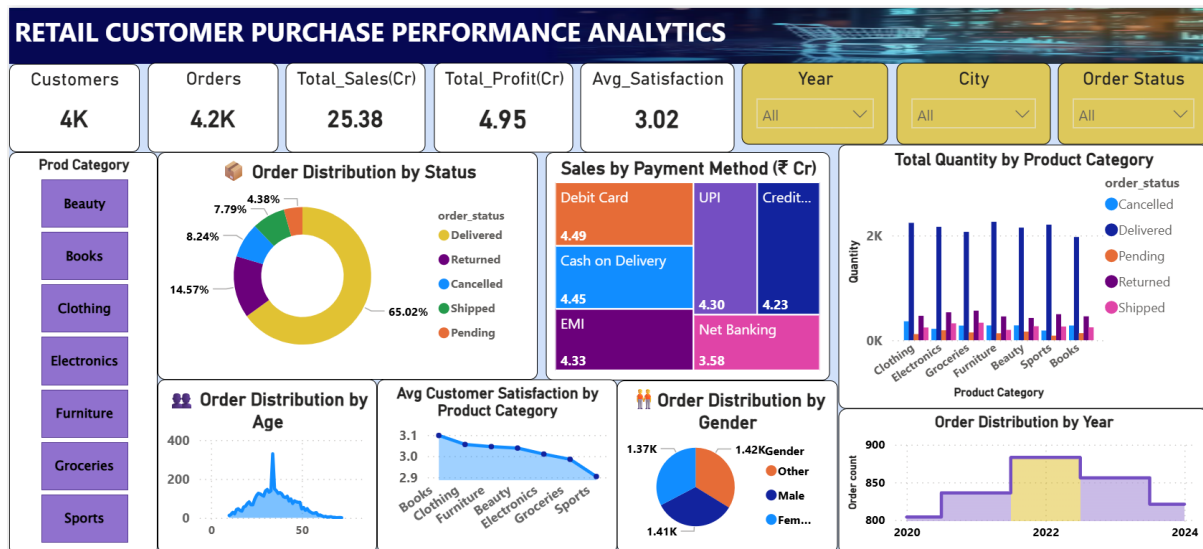


Average Sales Amount by Payment Method





DashBoard



Major Insights:

KPI Cards:

1. Customers

Total : 4K

This KPI represents the total number of unique customers in the dataset. The business has approximately 4,000 customers. Helps management understand the size of the customer base and track customer growth.

2. Orders

Total - 4.2K

This represents the **total number of orders** placed. Approximately **4,200 orders** were recorded. Since there are 4K customers and 4.2K orders, some customers have placed more than one order.

Useful for monitoring purchase activity and repeat-order behavior.

3. Total Sales (Cr)

Total - 25.38

This represents the **total sales generated**, expressed in crores. The business generated approximately **₹25.38 crore in sales**.

Provides the primary measure of overall revenue performance.

4. Total Profit (Cr)

Total - 4.95

This represents the **total profit generated**, expressed in crores. The business generated approximately **₹4.95 crore in profit**.

Helps management evaluate profitability rather than looking at sales alone.

The approximate profit margin based on these two KPIs is:

$$\text{₹4.95 Cr} \div \text{₹25.38 Cr} \times 100 \approx 19.5\%$$

Order Distribution by Gender

Key Insights:

- **Other:** 1.42K orders — the largest segment.
- **Male:** 1.41K orders.
- **Female:** 1.37K orders.
- Order distribution is **fairly balanced across all three gender groups**, with only a small difference between the highest and lowest segments.
- No single gender group dominates the overall order volume.

Business Recommendations

1. Use **inclusive marketing strategies** rather than focusing heavily on one gender segment.
2. Analyze **gender along with age, product category, and city** to identify more meaningful customer segments.
3. Compare **average order value and profit by gender** to determine whether purchasing volume translates into higher revenue or profitability.
4. Develop **personalized product recommendations** based on actual purchase behavior rather than gender alone.
5. Monitor gender-wise **customer satisfaction and return rates** to identify differences in customer experience.

Order Distribution by Status

Key Insights:

- **Delivered:** 65.02% — the majority of orders are successfully delivered.
- **Returned:** 14.57% — a significant portion of orders are returned.
- **Cancelled:** 8.24% of orders are cancelled.
- **Shipped:** 7.79% are currently in the shipped stage.
- **Pending:** 4.38% remain pending.
- Overall, 22.81% of orders are either returned or cancelled, which is an important area for further analysis.

Business Recommendations

1. **Reduce returns:** Identify products and categories with high return rates and investigate the underlying reasons.
2. **Analyze cancellations:** Examine whether cancellations are associated with specific products, cities, payment methods, or delivery delays.
3. **Improve delivery conversion:** Monitor the **Shipped** → **Delivered** transition to identify delivery bottlenecks.
4. **Address pending orders:** Investigate pending orders to determine whether payment, inventory, or logistics issues are causing delays.
5. **Track status by category:** Compare order status across product categories to identify categories with unusually high returns or cancellations.

Sales by Payment Method (₹ Cr)

Key Insights:

- **Debit Card:** ₹4.49 Cr — highest sales among the payment methods.
- **Cash on Delivery:** ₹4.45 Cr — very close to Debit Card.
- **EMI:** ₹4.33 Cr.
- **UPI:** ₹4.30 Cr.
- **Credit Card:** ₹4.23 Cr.
- **Net Banking:** ₹3.58 Cr — lowest sales contribution.
- Sales are **fairly well distributed** across most payment methods, with Net Banking showing the largest gap.

Business Recommendations

1. **Maintain multiple payment options** because customers are using a wide range of payment methods.
2. **Strengthen digital payment adoption**, particularly UPI and card payments, through convenient checkout experiences and targeted offers.
3. **Investigate Net Banking's lower contribution** to determine whether usability, availability, or customer preference is affecting adoption.
4. Analyze payment methods against **cancellation and return rates** to identify whether certain payment modes are associated with order issues.
5. Compare **profitability by payment method**, because the payment method with the highest sales may not necessarily generate the highest profit.

Total Quantity by Product Category & Order Status

Key insights:

- Furniture has the highest overall quantity/order volume among the displayed categories.

- Clothing, Sports, and Beauty also show relatively high volumes.
- Books has the lowest overall volume among the categories shown.
- Delivered orders are significantly higher than the other order statuses across all product categories.
- Returned orders form the next noticeable status segment, while Pending, Cancelled, and Shipped contribute comparatively smaller volumes.
- The consistently high Delivered volume indicates that most orders progress successfully through fulfillment.

Business Recommendations

1. **Maintain inventory for high-volume categories**
Ensure sufficient stock for categories such as Furniture, Clothing, Sports, and Beauty.
2. **Investigate returned orders**
Since Returned is the second-largest status in several categories, identify products with unusually high return volumes.
3. **Improve order fulfillment**
Analyze the gap between **Shipped and Delivered** orders to identify possible logistics delays.
4. **Monitor low-volume categories**
Investigate Books and other lower-volume categories to understand whether the issue is demand, product availability, or customer preference.
5. **Analyze category profitability**
High quantity does not necessarily mean high profit. Compare **Quantity → Sales → Profit → Return Rate** for each category.

Order Distribution by Age

Key Insights

- Orders are concentrated mainly among customers in the 20–50 age range.
- The distribution reaches its highest point around the mid-30s, indicating stronger order activity in this age group.
- Order volume gradually decreases after the 40s, with relatively fewer orders from older age groups.
- Very young and older customer segments contribute comparatively fewer orders.
- The sharp peak around the mid-30s indicates a particularly active customer age segment.

Business Recommendations

1. **Target the 30–40 age group**
Develop personalized promotions and product recommendations for the age group showing the highest order activity.
2. **Create age-based customer segments**
Compare **sales, profit, average order value and product preferences** across different age groups.
3. **Explore younger customers**
Use digital campaigns and introductory offers to increase engagement among younger age groups.
4. **Improve engagement among older customers**
Analyze their preferred products, payment methods and purchasing behavior before designing targeted campaigns.

5. **Combine age with product category**

Identify which products are most popular within each age segment to support targeted marketing and inventory planning.

Average Customer Satisfaction by Product Category

Key Insights

- **Books** has the highest average customer satisfaction at approximately **3.10**.
- Satisfaction gradually decreases across the categories.
- **Sports** has the lowest satisfaction at approximately **2.91**.
- **Clothing, Furniture, and Beauty** have relatively similar satisfaction levels around **3.04–3.06**.
- **Electronics and Groceries** are slightly below the 3.0–3.1 range.
- The gap between the highest and lowest categories is approximately **0.19 points**, indicating moderate variation across categories.

Business Recommendations

1. **Investigate Sports category**
 - Identify products receiving lower satisfaction.
 - Analyze returns, complaints, product quality and delivery experience.
2. **Learn from Books**
 - Identify factors contributing to higher satisfaction, such as product quality, delivery time or customer expectations.
 - Apply successful practices to lower-performing categories where relevant.
3. **Analyze satisfaction with returns**
 - Compare customer satisfaction against **return rates** by category.
 - Categories with both low satisfaction and high returns should receive priority.
4. **Analyze shipping experience**
 - Compare satisfaction with **average shipping days**.
 - Determine whether longer delivery times are associated with lower satisfaction.
5. **Improve product-level experience**
 - Drill down from category → product → satisfaction to identify specific products affecting the category average.

Order Distribution by Year

Key Insights

- Order volume increased from approximately **805 orders in 2020** to around **835 in 2021**.
- Orders reached their highest level at approximately **885 in 2022**.
- Order volume declined to around **855 in 2023**.
- Orders declined further to approximately **820 in 2024**.
- Overall, the chart shows **growth from 2020 to 2022, followed by a decline through 2024**.
- The 2022 peak indicates the strongest order activity during the period shown.

Business Recommendations

1. **Investigate the 2022 growth**

- Identify which products, cities, and customer segments contributed to the increase.
- Check whether promotions or specific product categories drove the growth.
- 2. **Analyze the post-2022 decline**
 - Compare **2022 vs. 2023–2024** by product category, city and customer segment.
 - Identify where order volume dropped the most.
- 3. **Improve customer retention**
 - Analyze repeat customers and customers who stopped purchasing after 2022.
 - Develop targeted retention campaigns.
- 4. **Focus on declining categories/cities**
 - Identify categories and cities contributing most to the decline.
 - Adjust product availability and marketing strategies accordingly.
- 5. **Compare orders with sales and profit**
 - A reduction in order count does not necessarily mean a reduction in business performance.
 - Analyze **Order Count + Sales + Profit + Average Order Value** together.

Sales by City (₹ Cr)

Key Insights

- The map shows that sales are distributed across multiple cities in India, rather than being concentrated in a single geographic region.
- Several cities in northern and central India show visible sales activity.
- Sales activity is also present across southern and eastern regions, indicating a broad geographic customer base.
- The bubble size represents the relative sales contribution of each city, allowing high- and low-performing locations to be identified geographically.
- The map is useful for identifying geographic sales patterns, but exact city-level comparisons are difficult from the map alone.

Business Recommendations

1. **Identify high-sales cities**
 - Use a ranked table or bar chart alongside the map to identify the exact top-performing cities.
2. **Strengthen high-performing markets**
 - Ensure adequate inventory, delivery capacity and customer support in cities generating higher sales.
3. **Investigate lower-performing cities**
 - Compare product demand, order volume, average order value and customer segments to understand lower sales performance.
4. **Develop city-specific strategies**
 - Customize product availability and marketing campaigns according to local purchasing patterns.
5. **Combine geography with profitability**
 - Don't evaluate cities using sales alone. Compare **Sales + Profit + Orders + Return Rate + Customer Satisfaction**.

Top 10 Products by Sales (₹ Cr)

Key Insights

- Laptop is the highest-selling product among the Top 10.
- Smartwatch and Headphones follow closely behind Laptop.
- Camera and Smartphone are also strong contributors to sales.
- Tablet has moderate sales compared with the leading products.
- Desk, Bed Frame, and Wardrobe have comparatively lower sales within the Top 10.
- The leading products are primarily electronics, indicating strong sales contribution from technology-related products.

Business Recommendations

1. **Maintain sufficient inventory** for high-performing products such as Laptop, Smartwatch and Headphones.
2. **Use cross-selling strategies** — for example, recommend headphones or accessories with laptops and smartphones.
3. **Analyze profitability**, not just sales, to determine which products generate the highest margins.
4. **Investigate lower-performing products** to understand whether pricing, demand, availability or promotions are affecting sales.
5. **Monitor product trends regularly** so inventory and marketing strategies can be adjusted based on changing customer demand.

Sales by Product Category & Order Status

Key Insights

- Electronics is the dominant category, generating substantially higher sales than the other categories.
- Furniture is the second-highest contributor to sales.
- Sports follows Furniture but at a considerably lower level.
- Clothing, Beauty, Books, and Groceries contribute comparatively smaller sales.
- Delivered orders account for the largest portion of sales across the categories.
- Returned, cancelled, pending and shipped sales are much smaller compared with delivered sales.

Business Recommendations

1. **Prioritize Electronics**
 - Maintain sufficient inventory for high-demand electronics.
 - Monitor stock-outs because this category contributes significantly to overall sales.
2. **Strengthen Furniture performance**
 - Analyze which furniture products drive sales and identify opportunities for cross-selling.
3. **Investigate lower-performing categories**
 - Review pricing, product availability, promotions and customer demand for Books, Groceries, Beauty and Clothing.
4. **Monitor non-delivered sales**
 - Analyze cancelled and returned orders within each category to identify potential revenue leakage.
5. **Compare sales with profit**
 - Electronics may have high sales, but management should also evaluate profit margin, discounts and shipping costs before prioritizing products.

Average Order Quantity by Category

Key Insights

- Furniture has the highest average order quantity at approximately 5.8 units per order.
- Electronics and Groceries follow closely at around 5.6–5.7 units.
- Books and Beauty have an average of approximately 5.5 units.
- Clothing is slightly lower at around 5.4 units.
- Sports has the lowest average order quantity at approximately 5.3 units.
- The difference between the highest and lowest categories is relatively small, indicating that average order quantity is fairly consistent across categories.

Business Recommendations

1. **Prioritize Furniture inventory**
Since Furniture has the highest average quantity per order, ensure adequate stock availability.
2. **Analyze high-volume categories**
Examine Furniture, Electronics and Groceries to identify opportunities for **bundle offers and bulk purchasing**.
3. **Investigate Sports category**
Analyze why Sports has the lowest average quantity and determine whether pricing, product type or customer behavior contributes to the difference.
4. **Use quantity with sales and profit**
A higher quantity does not necessarily mean higher revenue or profit. Compare:
Average Quantity → Sales → Profit → Profit Margin
5. **Create product bundles**
Categories with higher average quantities may provide opportunities for bundle or multi-item promotions.

Total Sales & Total Profit by Year

Key Insights

- **2020:** Sales and profit start at relatively lower levels.
- **2021:** Sales increase to approximately ₹5.4 Cr, while profit also rises significantly.
- **2022:** Sales remain around ₹5.35 Cr, but profit declines compared with 2021.
- **2023:** Sales fall noticeably to approximately ₹4.8 Cr, while profit reaches its **highest** level in the period.
- **2024:** Sales recover to approximately ₹5.0 Cr, while profit decreases from the 2023 peak.
- The chart shows that **higher sales do not always correspond to higher profit**. The 2023 results are a good example: sales are lower, but profit is higher.

Business Recommendations

1. **Investigate 2023 profitability**
 - Identify why profit increased despite lower sales.
 - Analyze discounts, product mix, shipping costs and operating costs.
2. **Analyze the 2023 sales decline**
 - Identify which cities, products and categories contributed to the reduction in sales.
3. **Focus on profitable products**

- Don't evaluate products based only on sales volume.
 - Identify products with strong **sales and profit margins**.
4. **Monitor profit margin annually**
 - Track **Sales** → **Profit** → **Profit Margin** together to understand business efficiency.
 5. **Investigate the 2024 profit decline**
 - Sales recovered in 2024, but profit decreased from 2023.
 - Review discounts, product mix, returns and shipping costs.

Average Shipping Days by Product Category

Key Insights

- **Beauty, Clothing, and Electronics** have the highest average shipping time at **6 days**.
- **Books, Furniture, Groceries, and Sports** have a lower average shipping time of **5 days**.
- The difference between the fastest and slowest categories is only **1 day**, indicating relatively consistent shipping performance.
- The higher shipping time for Beauty, Clothing, and Electronics may be worth investigating, particularly if these categories also have lower customer satisfaction.

Business Recommendations

1. **Investigate the 6-day categories**
Analyze logistics, warehouse processing and delivery locations for Beauty, Clothing and Electronics.
2. **Compare shipping time with satisfaction**
Check whether the additional day of shipping is contributing to lower customer satisfaction.
3. **Optimize logistics**
Identify opportunities to reduce processing and delivery time for the 6-day categories.
4. **Analyze city-level performance**
A category may have a 6-day average because of specific cities or regions. Drill down into **City** → **Category** → **Shipping Days**.
5. **Set delivery targets**
Establish category-wise delivery targets and monitor actual performance against them.

9. Final Overall Insights

Descriptive Analysis

- 4K customers and 4.2K orders were recorded.
- Total sales reached ₹25.38 Cr with ₹4.95 Cr profit. 65.02% of orders were delivered. 14.57% of orders were returned and 8.24% were cancelled.
- Electronics generated the highest category-level sales.
- Laptop was the top-selling product among the Top 10.
- Average customer satisfaction was 3.02/5.

Diagnostic Analysis

- High return and cancellation rates require further investigation.
- Sports recorded the lowest average customer satisfaction.

- Beauty, Clothing and Electronics had relatively higher 6-day shipping times.
- Sales declined after the 2022 peak, requiring analysis by product, city and customer segment.
- Profit performance varied independently from sales, indicating the need to examine product mix, discounts and costs.
- Net Banking recorded the lowest sales among the payment methods shown.

Predictive Analysis

- Monitor the decline in order volume after 2022.
- High-performing products can be monitored for future demand and inventory requirements.
- Lower-satisfaction categories may require monitoring for potential effects on returns and repeat purchases.
- Historical city and category performance can be used as inputs for future sales and demand forecasting.

Prescriptive Analysis

- Reduce returns by identifying high-return products and categories.
- Improve customer satisfaction by addressing issues in lower-performing categories.
- Maintain sufficient inventory for high-sales products, particularly Electronics.
- Optimize logistics for categories with higher shipping times.
- Monitor sales, profit, discounts, shipping costs and returns together to improve profitability.
- Develop targeted customer strategies using age, city, product preference and purchase behavior.

10. Conclusion:

- The analysis provides a comprehensive view of customer behavior, sales performance, profitability, order trends, product performance, and shipping efficiency.
- The business generated ₹25.38 Cr in sales and ₹4.95 Cr in profit from approximately 4K customers and 4.2K orders.
- Electronics and products such as Laptops, Smartwatches, and Headphones are major contributors to sales.
- While 65.02% of orders were delivered, the 14.57% return rate and 8.24% cancellation rate highlight opportunities to improve order fulfillment.
- Customer satisfaction averaged 3.02/5, with differences across product categories that can help identify areas for customer-experience improvement.
- Order and sales trends show changes across years, providing useful information for understanding business performance and planning future strategies.
- Overall, the insights can support data-driven decisions related to inventory, customer retention, product strategy, logistics, and profitability improvement.